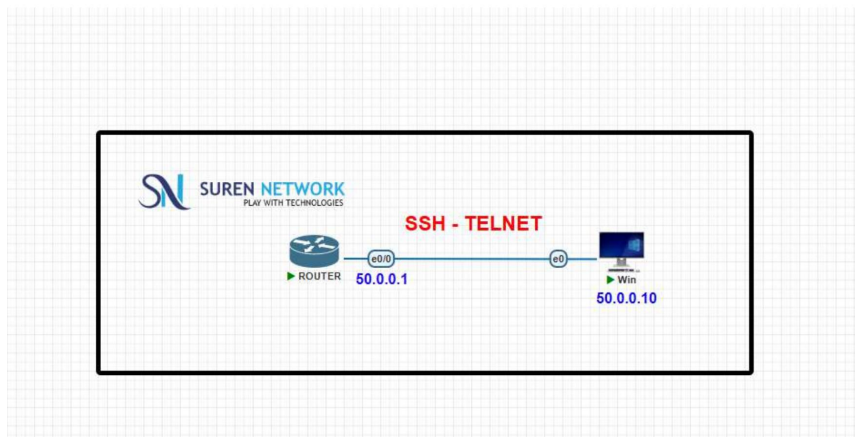


CCNA Lab Report

Router Configuration & Introduction to Telnet



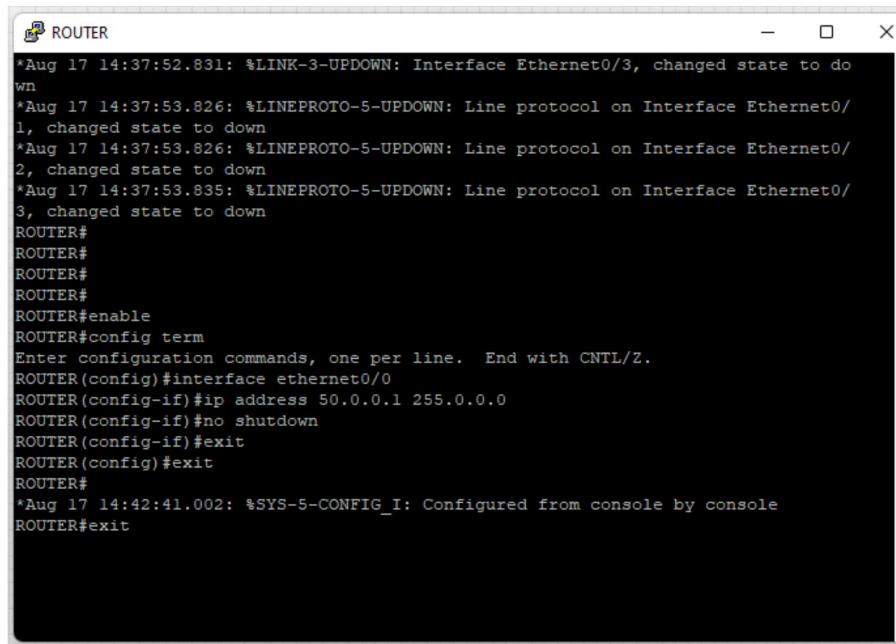
Lab Topology — Router (50.0.0.1/8) ↔ Windows PC (50.0.0.10/8)

Objective	Configure a router's Ethernet interface, verify PC-router connectivity, and set up local/console authentication as groundwork for Telnet access.
Tool	Cisco IOS Router + Windows PC (EVE-NG / Packet Tracer lab)
Topic	Router Configuration and Intro to Telnet – CCNA

Step 1 | Configure the Router Interface

The router boots with all interfaces down. Ethernet0/0 is assigned an IP address in the 50.0.0.0/8 network and brought up with **no shutdown**.

```
ROUTER#enable
ROUTER#config term
ROUTER(config)#interface ethernet0/0
ROUTER(config-if)#ip address 50.0.0.1 255.0.0.0
ROUTER(config-if)#no shutdown
ROUTER(config-if)#exit
ROUTER(config)#exit
```



Router console: link-down messages on unused interfaces, followed by Ethernet0/0 configuration.

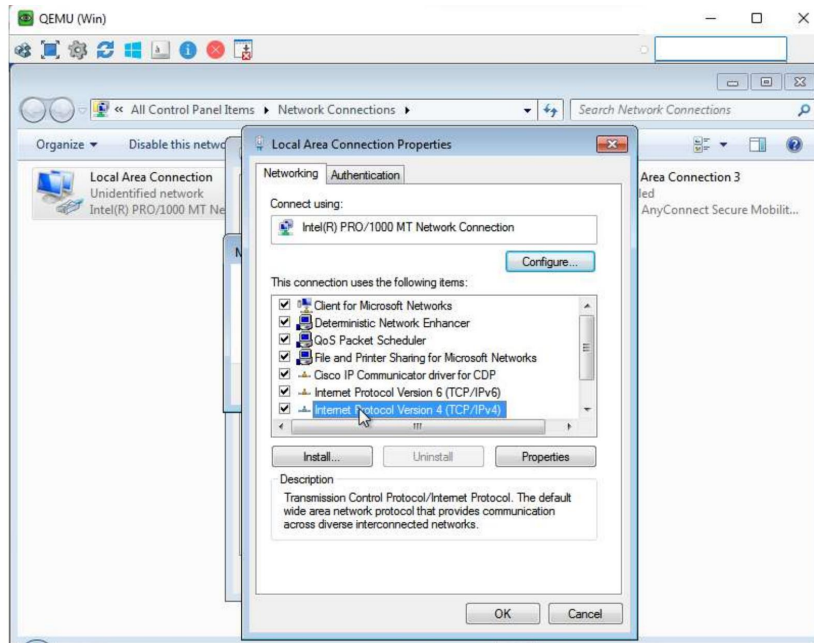
Step 2 | Configure the Windows PC IP Address

On the Windows PC, **ncpa.cpl** is run from Command Prompt to open Network Connections directly.

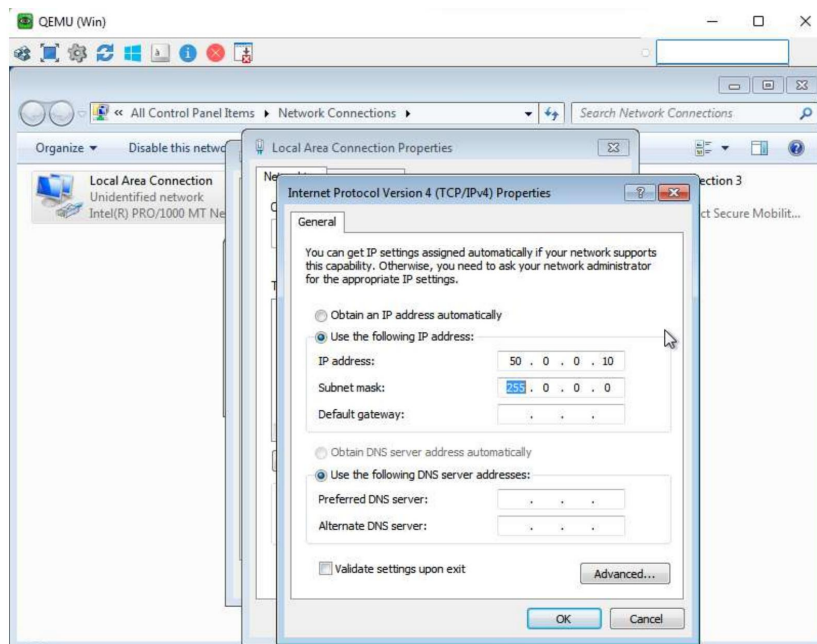


Running ncpa.cpl to jump straight to Network Connections.

The Local Area Connection's **IPv4 Properties** are opened to set a static address.



Network Connections → Local Area Connection Properties → TCP/IPv4.



Static IP configured: 50.0.0.10 / 255.0.0.0 (no gateway needed — same subnet).

Verified with **ipconfig**:

```
C:\Users\user>ipconfig
```

```
Ethernet adapter Local Area Connection:
```

```
IPv4 Address. . . . . : 50.0.0.10
```

```
Subnet Mask . . . . . : 255.0.0.0
```

```
Administrator: Command Prompt
C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b020:d0a:5681:8ebb%11
    IPv4 Address. . . . . : 50.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{42A1F8B7-30A8-4EF7-81AF-C694F5F3D81B}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter 6T04 Adapter:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2002:3200:a::3200:a
    Default Gateway . . . . . : 

C:\Users\user>_
```

ipconfig output confirming the assigned IPv4 address.

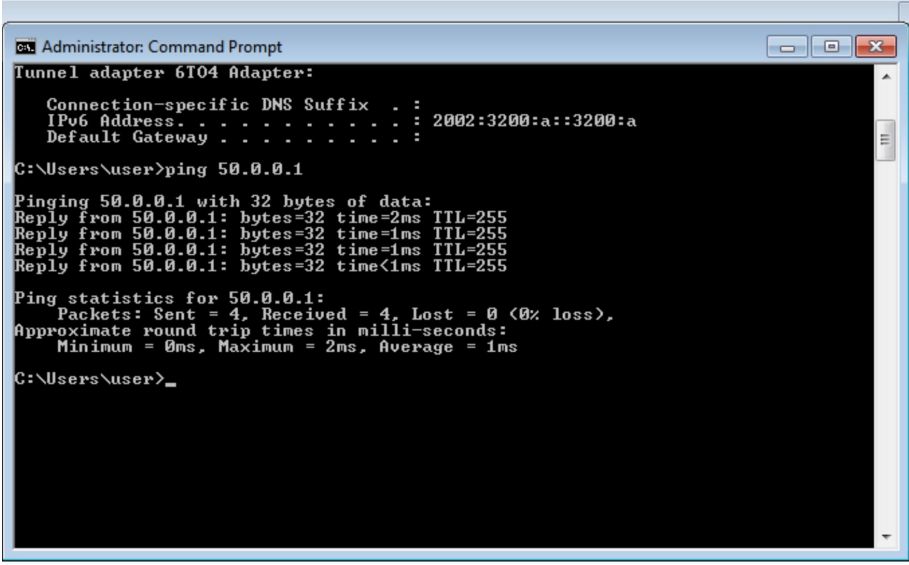
Step 3 | Verify Connectivity

A ping from the PC to the router confirms Layer 3 reachability:

```
C:\Users\user>ping 50.0.0.1

Reply from 50.0.0.1: bytes=32 time=2ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```



```
Administrator: Command Prompt
Tunnel adapter 6T04 Adapter:
    Connection-specific DNS Suffix . . : 
    IPv6 Address. . . . . : 2002:3200:a::3200:a
    Default Gateway . . . . . : 

C:\Users\user>ping 50.0.0.1

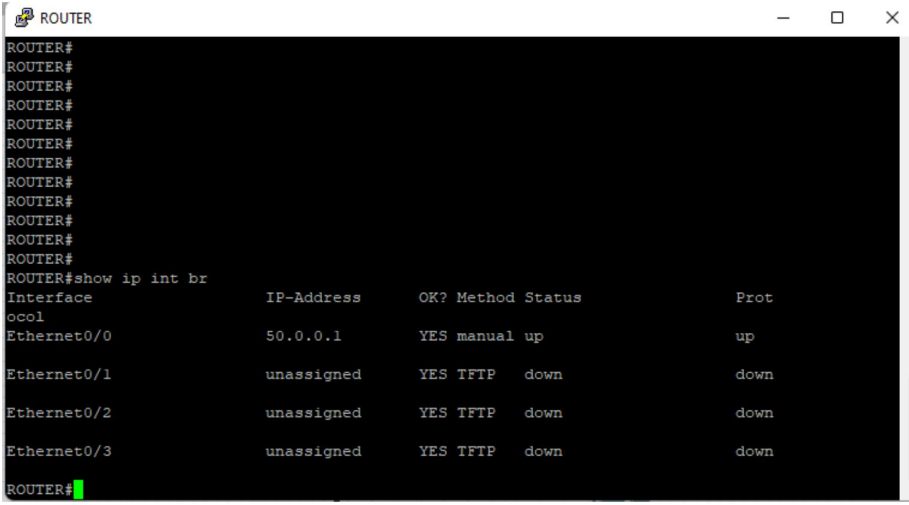
Pinging 50.0.0.1 with 32 bytes of data:
Reply from 50.0.0.1: bytes=32 time=2ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Users\user>
```

Successful ping — 0% packet loss between PC and router.

Confirmed on the router side with **show ip int br**:



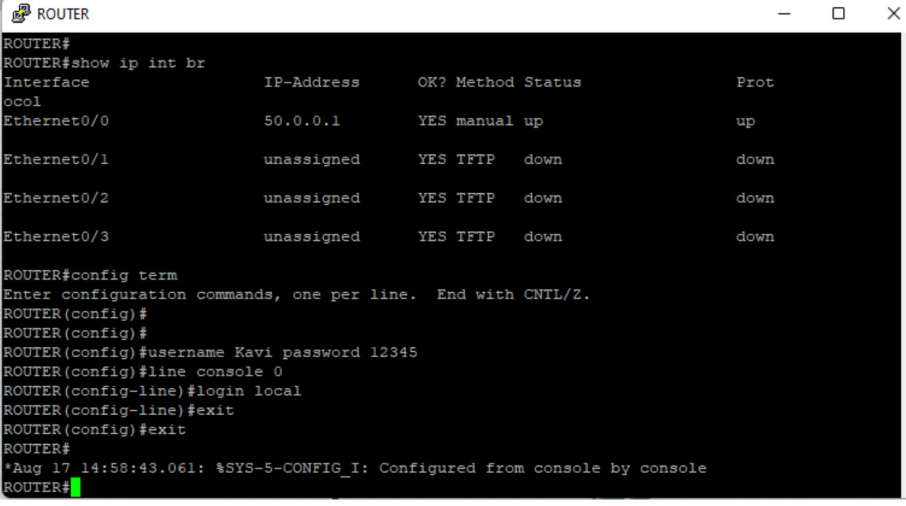
```
ROUTER
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#
ROUTER#show ip int br
Interface              IP-Address      OK? Method Status      Prot
Ethernet0/0            50.0.0.1       YES manual up           up
Ethernet0/1            unassigned     YES TFTP  down        down
Ethernet0/2            unassigned     YES TFTP  down        down
Ethernet0/3            unassigned     YES TFTP  down        down
ROUTER#
```

Ethernet0/0 shows up/up with IP 50.0.0.1; unused interfaces remain down.

Step 4 | Local Username & Console Login

A local user account is created and the console line is set to authenticate against it using **login local**.

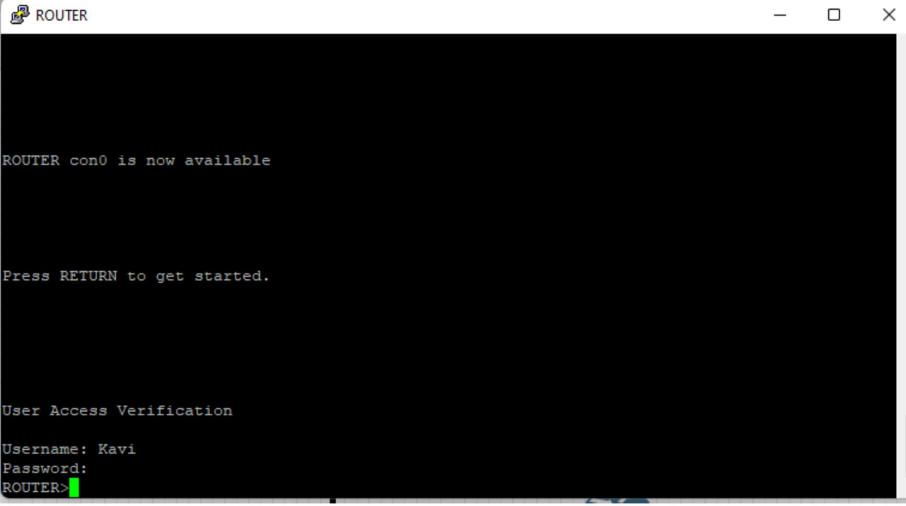
```
ROUTER(config)#username Kavi password 12345
ROUTER(config)#line console 0
ROUTER(config-line)#login local
ROUTER(config-line)#exit
ROUTER(config)#exit
```



```
ROUTER#
ROUTER#show ip int br
Interface      IP-Address      OK? Method Status  Prot
-----
Ethernet0/0    50.0.0.1        YES manual up      up
Ethernet0/1    unassigned      YES TFTP  down    down
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down

ROUTER#config term
Enter configuration commands, one per line. End with CNTL/Z.
ROUTER(config)#
ROUTER(config)#
ROUTER(config)#username Kavi password 12345
ROUTER(config)#line console 0
ROUTER(config-line)#login local
ROUTER(config-line)#exit
ROUTER(config)#exit
ROUTER#
*Aug 17 14:58:43.061: %SYS-5-CONFIG_I: Configured from console by console
ROUTER#
```

Local username 'Kavi' created and console line set to login local.



```
ROUTER con0 is now available

Press RETURN to get started.

User Access Verification

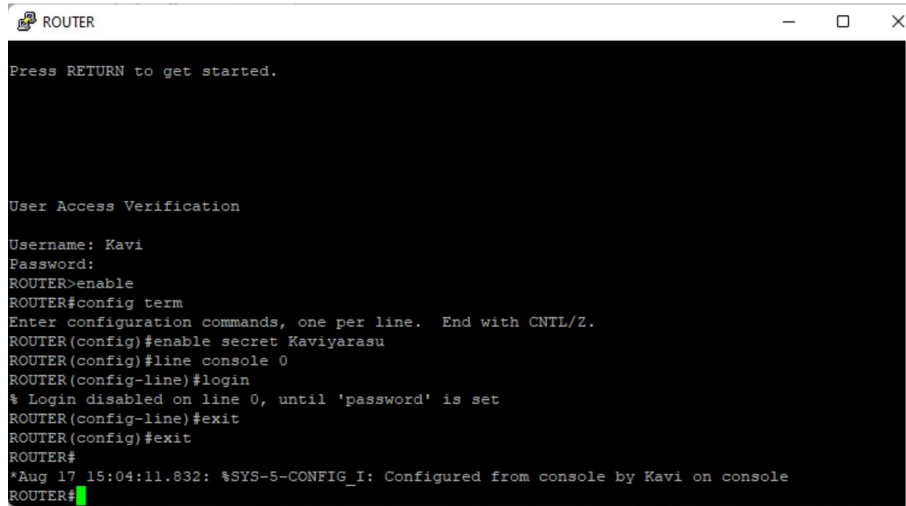
Username: Kavi
Password:
ROUTER>
```

Console re-prompts for the local username/password before granting access.

Step 5 | Enable Secret Password

An **enable secret** (encrypted) password is set to protect privileged EXEC mode.

```
ROUTER(config)#enable secret Kaviyarasu
```



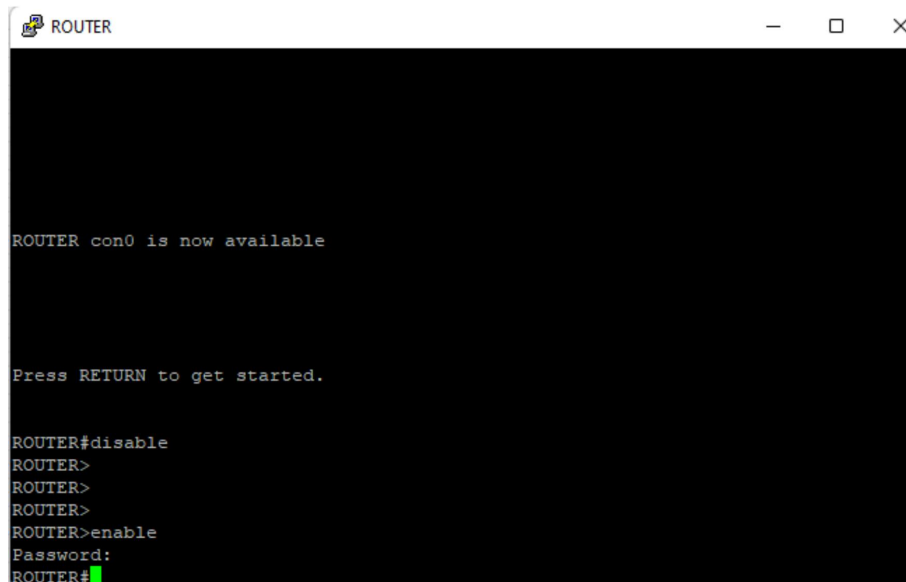
```
ROUTER
Press RETURN to get started.

User Access Verification

Username: Kavi
Password:
ROUTER>enable
ROUTER#config term
Enter configuration commands, one per line. End with CNTL/Z.
ROUTER(config)#enable secret Kaviyarasu
ROUTER(config)#line console 0
ROUTER(config-line)#login
% Login disabled on line 0, until 'password' is set
ROUTER(config-line)#exit
ROUTER(config)#exit
ROUTER#
*Aug 17 15:04:11.832: %SYS-5-CONFIG_I: Configured from console by Kavi on console
ROUTER#
```

enable secret configured; entering enable now prompts for this password.

Note: attempting **login** (without **local**) on *line console 0* without a line password first produced the expected IOS message: % Login disabled on line 0, until 'password' is set. This is normal — plain **login** requires a **password** to be set on that line, unlike **login local** which uses the username database instead.



```
ROUTER
ROUTER con0 is now available

Press RETURN to get started.

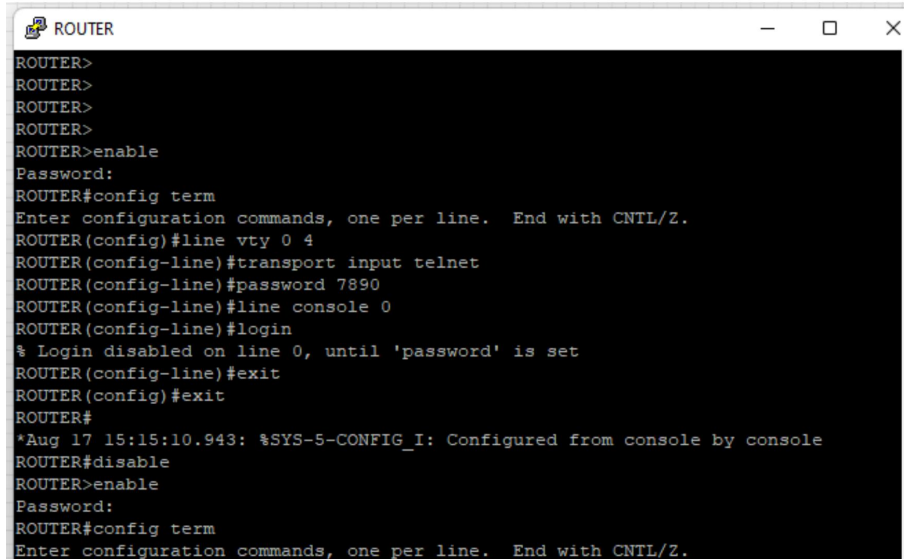
ROUTER#disable
ROUTER>
ROUTER>
ROUTER>
ROUTER>enable
Password:
ROUTER#
```

Testing privileged mode: disable → enable now asks for the enable password.

Step 6 | Prepare VTY Lines for Telnet

Finally, the VTY (virtual terminal) lines are configured to accept incoming Telnet connections with a password, laying the groundwork for remote access in the next stage of the lab.

```
ROUTER(config)#line vty 0 4
ROUTER(config-line)#transport input telnet
ROUTER(config-line)#password 7890
```



```
ROUTER>
ROUTER>
ROUTER>
ROUTER>
ROUTER>enable
Password:
ROUTER#config term
Enter configuration commands, one per line. End with CNTL/Z.
ROUTER(config)#line vty 0 4
ROUTER(config-line)#transport input telnet
ROUTER(config-line)#password 7890
ROUTER(config-line)#line console 0
ROUTER(config-line)#login
% Login disabled on line 0, until 'password' is set
ROUTER(config-line)#exit
ROUTER(config)#exit
ROUTER#
*Aug 17 15:15:10.943: %SYS-5-CONFIG_I: Configured from console by console
ROUTER#disable
ROUTER>enable
Password:
ROUTER#config term
Enter configuration commands, one per line. End with CNTL/Z.
```

VTY lines 0-4 configured for Telnet with password 7890.

Key Learnings

- **no shutdown** is required to bring an interface administratively up.
- Both ends of a link need matching subnet/mask for Layer 3 connectivity.
- **ping** and **show ip int br** are the go-to tools to verify reachability.
- **username + login local** enables local-database authentication on a line.
- **enable secret** stores an encrypted privileged-mode password (preferred over **enable password**).
- **line vty 0 4 + transport input telnet + password** is the minimum setup for Telnet remote access.